GO-6 — Physical-AI Benchmark

A multimodal technical-decision benchmark based on analog instruments, fuzzy logic, and dynamic simulation

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This document formalizes GO-6 as a new benchmark class: not merely a test of formal calculation, but a test of perception, measurement intervals, fuzzy risk assessment, simulation of action consequences, and executable programmatic verification.

Abstract

GO-6 extends the GO benchmark philosophy from purely text-formal tasks into the domain of physical, multimodal, and decision-oriented reasoning. The test was designed because we wanted to determine whether an LLM/LMM system can operate through a more realistic engineering chain: reading analog instruments, accepting measurement uncertainty, performing a fuzzy assessment of system state, simulating system dynamics over multiple steps, classifying the consequences of each control action, selecting a uniquely best decision, and then expressing the same procedure as verifiable Zig code and structured JSON output. Unlike standard digital tasks in which input values are already ideally prepared, GO-6 deliberately introduces physical imprecision, transitional fuzzy zones, and a conflict between an intuitively aggressive reaction and the actual dynamic consequences of that reaction. It therefore measures whether a model can reason through perception, context, causal propagation, and technical verifiability, instead of merely producing persuasive text.

Keywords: physical-AI benchmarking, LMM, analog instruments, fuzzy logic, dynamic simulation, technical decision-making, Zig verification, REAL-AI-Benchmark.

Executive summary

GO-6 introduces analog instruments as the input layer, fuzzy inference as the layer for assessing current risk, and discrete dynamic simulation as the layer for choosing a control action. The principal methodological purpose is to separate a model that merely recognizes a pattern from a model that can read physical quantities, reason with intervals, avoid premature decisions, and understand that the most aggressive action is not necessarily the best one. The test is compatible with two versions: GO-6-old-simple-no-VI, in which the inputs are provided digitally for ordinary text-only LLMs, and the main GO-6 multimodal version, in which the model must read five images of analog instruments.

1. Introduction: why GO-6 was necessary

GO-1 through GO-5 already cover formal logic, exact numerical reasoning, manual convolution, bitmask search, a Mealy transducer, proof of uniqueness, and executable code. However, all of those tasks begin from a text in which the input values are already provided in a clean digital form. This is necessary for testing formal discipline, but it does not cover the full space of real technical decision-making.

For that reason, GO-6 deliberately moves from the world of perfectly prepared symbols into the world of physical perception. The aim was not to create one more mathematical exercise, but to check clearly whether a model can do what an engineer or operator must do in a real system: look at an instrument, read a value, acknowledge that the reading has a tolerance, assess the system state, and only then compute the consequences of possible interventions.

In such a chain, an error is not born only in arithmetic. It may arise already at the perceptual stage. A model may misread a scale, ignore an interval, equate a fuzzy warning with the final control decision, skip dynamic consequences, or write code that does not reproduce its own analysis. GO-6 is therefore introduced as a physical-multimodal extension of the benchmark, not as a cosmetic variant of the previous tests.

2. Methodological position of GO-6 relative to GO-1 through GO-5

GO-6 does not change the basic philosophy of the GO set: intermediate steps, formal structure, code, JSON, and provability are still assessed. What changes is the nature of the input and the nature of the decision. The model no longer receives only a textual problem; it must translate visual-physical information into a numerical model.

Test layer	GO-1 to GO-5	GO-6-old-simple-no-VI	GO-6 multimodal test
Input	Digitally specified text, matrices, and formulas	Digitally specified physical quantities	Images of analog instruments + textual rules
Main error risk	Formal calculation, logic, code	Fuzzy calculation and dynamics	Reading, interval, fuzzy logic, dynamics, and code
Multimodality	Not required	Not required	Required
Final signal	Correct solution and code	Correct physical decision	Correct decision derived from perception and simulation

Table 1. Comparison of GO-6 with the preceding benchmark layers.

3. Architecture of the GO-6 test

GO-6 is organized as a controlled engineering problem in which a system is in a stressed, but not trivially catastrophic, state. Five physical quantities are read from analog instruments: temperature T, pressure P, flow F, vibration V, and pressure trend D = dP/dt. These values are used to compute a fuzzy risk score, after which four possible control actions are simulated.

Symbol	Quantity	Role in the system
T	Temperature [°C]	Thermal state; affected by pressure, flow, and the action heat term H.
P	Pressure [kPa]	Mechanical load; affected by trend D, temperature, and pressure relief R.
F	Flow [L/min]	Cooling and transport; changed by action C and influences temperature.
V	Vibration [mm/s]	Dynamic-mechanical channel; decisive for unstable/catastrophic boundaries.
D	dP/dt [kPa/min]	Pressure trend; indicates whether the system is naturally moving toward worsening or stabilization.

Table 2. Physical quantities in the GO-6 test and their roles.

4. Analog instruments as a deliberate methodological filter

Analog instruments were introduced because, in a real physical system, measurement is not an ideal JSON object. A pointer on a scale requires interpolation, and interpolation requires an interval. A model that reports only a single digital value without an interval has in fact skipped an important part of the physical problem.

In the reference GO-6 instance, the images are named PIC1.png through PIC5.png. In this document they are shown as illustrative instruments; in the actual test, the images are attached manually to the prompt while the textual task statement remains fixed.

PIC1: T	PIC2: P	PIC3: F	PIC4: V	PIC5: D
68.5 °C	132.0 kPa	4.2 L/min	5.6 mm/s	+2.1 kPa/min
[68.0, 69.0]	[131.0, 133.0]	[4.0, 4.4]	[5.4, 5.8]	[1.9, 2.3]

Figure 1. Reference analog instruments for the GO-6 test. The images in this document provide the working visual basis for the multimodal version of the benchmark.

5. Physical equations and the dynamic model

The dynamic model is intentionally small, but causally dense. Each action has four parameters: H for thermal effect, R for pressure relief, S for vibration shock, and C for flow change. The state is updated over three ticks, and each tick uses the previous state. Consequently, calculation errors propagate through the simulation.

```
T_next = T + 0.18 * (P - 120) - 0.75 * F + H
P_next = P + D + 0.12 * (T - 65) - R
V_next = V + 0.08 * (P - 125) + S
F_next = F + C
```

Action	Name	H	R	S	C
A0	MONITOR	+2.0	0.0	+0.2	0.0
A1	DERATE	+0.5	1.0	+0.1	+0.4
A2	CONTROLLED_RELIEF	-0.5	4.0	+0.3	+0.8
A3	EMERGENCY_MODE	-2.0	9.0	+1.8	-1.5

Table 3. Control-action parameters in the GO-6 model.

6. Fuzzy inference: risk assessment, not the final decision

The fuzzy layer is introduced because physical quantities in transitional regions are not naturally binary. A temperature may be partially normal and partially high; pressure may lie in the transition from normal to high; vibration may already be unstable, while still not catastrophic. Therefore, membership degrees and Mamdani min-max rules are used.

For the reference nominal values, the key membership degrees are:

Quantity	Fuzzy set	Membership degree
T	normal	0.100
T	high	0.900
T	critical	0.000
P	normal	0.150
P	high	0.850
P	critical	0.000
F	low	0.400
F	adequate	0.600
V	stable	0.000
V	unstable	0.640
D	steady	0.000
D	rising_fast	0.550

Table 4. Reference fuzzy memberships for the nominal GO-6 readings.

Rule activations are computed using min-max logic:

```
R1 = min(high(T), high(P))        = min(0.900, 0.850) = 0.850
```

```
R2 = min(high(P), rising_fast(D)) = min(0.850, 0.550) = 0.550
R3 = min(low(F), high(T))         = min(0.400, 0.900) = 0.400
R4 = min(unstable(V), high(P))    = min(0.640, 0.850) = 0.640
R5 = max(critical(T), critical(P)) = 0.000

aggregated_risk = max(R1, R2, R3, R4, R5) = 0.850
initial_fuzzy_decision = EMERGENCY_MODE
```

The key methodological point is that this decision is not the final action. It is only an assessment of the current risk. The final control action must be derived from simulation of consequences, because an aggressive intervention may introduce secondary risk.

7. Simulation of consequences and action selection

GO-6 deliberately separates warning from control. If risk is high, this does not imply that the most aggressive action is the best action. In the reference scenario, EMERGENCY_MODE is intuitively tempting, yet it produces catastrophic vibration.

Action	T3	P3	F3	V3	Outcome	max rel. excess
A0 MONITOR	72.913	139.978	4.200	8.494	catastrophic	0.213499
A1 DERATE	66.940	136.380	5.400	7.940	unstable	0.134299
A2 CONTROLLED_RELIEF	61.399	126.920	6.600	7.810	unstable	0.115785
A3 EMERGENCY_MODE	59.341	111.479	-0.300	11.096	catastrophic	1.085714

Table 5. Nominal dynamic outcomes after three ticks.

According to the selection criterion, catastrophic actions are eliminated first. The procedure then looks for a safe action. If no safe action exists, it selects the unstable action with the smallest maximum relative excess above the safety boundaries. In the reference case, A1 and A2 are both unstable, but A2 has the smaller relative vibration excess above the V = 7 boundary.

```
excess(A1) = (7.940096 - 7) / 7 = 0.134299
excess(A2) = (7.810496 - 7) / 7 = 0.115785

0.115785 < 0.134299
therefore recommended_action = CONTROLLED_RELIEF
```

8. What GO-6 validates

Validation layer	What is measured
Visual reading	The model must identify the scale, pointer position, and approximate value, with a reasonable uncertainty interval.
Measurement uncertainty	The benchmark does not reward false precision; it rewards a technically reasonable combination of nominal value and reading interval.
Fuzzy reasoning	The model must compute with membership degrees and min-max rules, not merely with hard if/else thresholds.
Causal dynamics	The model must propagate the state through three ticks and must not reuse stale values in later steps.
Decision-making under conflict	The model must recognize that the highest initial risk does not automatically justify the most aggressive action.
Code verifiability	The Zig code must reproduce the fuzzy and dynamic layers, not merely print a known result.

Table 6. Validation layers of the GO-6 test.

9. Typical model errors and expected failure modes

GO-6 is constructed to reveal several weaknesses that often remain hidden in broad but weakly controlled benchmark collections. The most common expected errors are the following:

Error class	Example	Effect on score
Incorrect reading	The pressure pointer is read as 128 kPa or 138 kPa instead of approximately 132 kPa.	Affects section A and propagates into C/D.

No interval	The model reports T = 68.5 without an interval or comment on analog uncertainty.	Reduces A and methodological maturity.
Fuzzy shortcut	The model directly selects EMERGENCY_MODE from aggregated_risk = 0.850.	Major error in E.
Simulation drift	At the second tick the model uses the initial P instead of P1, or fails to update F.	Major error in D.
Ignoring vibration	The model treats A3 as good because it lowers temperature and pressure, but misses V3 > 8.	Critical decision error.
Non-executable Zig	The code is pseudocode or uses an incompatible I/O pattern for the target Zig version.	Loss of F-section points.

Table 7. Typical errors and their interpretation.

10. Proposed scoring rubric

GO-6 should be scored by summing achievements across elements. There is no arbitrary rule that one failed element automatically collapses the whole answer to a low score. If a model is excellent in perception and decision-making but has minor Zig issues, it loses points from the code component. If the code runs, but the final decision is wrong, the code cannot save the overall score.

Section	Scored element	Points / 100
A	Reading five analog instruments, nominal values, intervals, and brief justification	15
B	Physical consistency and understanding of conflict between system channels	10
C	Fuzzy memberships, rule activations, aggregated_risk, and initial fuzzy decision	20
D	Dynamic simulation of all four actions across three ticks and correct outcome classification	20
E	Final decision, selection of A2, proof of uniqueness, and explanation of why the fuzzy decision is not final	15
F	Zig code: algorithm, compatibility, JSON output, and result reproduction	15
G	Format, sections, RESULTS_JSON, and final <END> marker	5

Table 8. Proposed GO-6 scoring rubric.

11. Compact reference Zig snippets

The snippets below are not a complete final program; they show the elements that the code must genuinely implement. A full model answer should implement the complete fuzzy functions, all actions, simulation, classification, selection, and RESULTS_JSON output.

```
const State = struct { T: f64, P: f64, F: f64, V: f64 };
const Action = struct { name: []const u8, index: i32, H: f64, R: f64, S: f64, C: f64 };

fn simulate(s0: State, dPdt: f64, a: Action) [3]State {
    var states: [3]State = undefined;
    var s = s0;
    for (0..3) |i| {
        const next = State{
            .T = s.T + 0.18 * (s.P - 120.0) - 0.75 * s.F + a.H,
            .P = s.P + dPdt + 0.12 * (s.T - 65.0) - a.R,
            .V = s.V + 0.08 * (s.P - 125.0) + a.S,
            .F = s.F + a.C,
        };
        states[i] = next;
        s = next;
    }
    return states;
}

fn maxRelativeExcess(s: State) f64 {
    return max4(
        max(0.0, (s.T - 78.0) / 78.0),
```

```
        max(0.0, (s.P - 142.0) / 142.0),
        max(0.0, (s.V - 7.0) / 7.0),
        max(0.0, (3.5 - s.F) / 3.5),
    );
}
```

12. GO-6-old-simple-no-VI as a control variant

The digital version of GO-6, designated GO-6-old-simple-no-VI, serves to test models without visual capabilities. In that version, the instruments are not attached as images; instead, the input values are provided directly in text. This version measures the fuzzy-dynamic decision process, but it does not measure visual reading. Therefore, its results should not be mixed with the multimodal results unless the version is explicitly marked. The multimodal version has a wider evaluative scope and should be treated as the main GO-6 test.

13. Proposed execution protocol

- Use the same task statement and the same instrument images PIC1.png through PIC5.png for all models in the main GO-6 comparison.
- For multimodal models, record whether the images were actually attached and whether the model explicitly read values from them.
- For text-only models, use GO-6-old-simple-no-VI and clearly mark the result as no-visual-input.
- Preserve the full raw answer, runtime metadata, tokens/s, total time, token count, and whether the Zig code compiles.
- Score additively according to the rubric, without changing the methodology retroactively for an individual model.
- Separate all-runs tables from best-by-model tables, as in GO-1 through GO-5.

14. Conclusion

GO-6 opens a new field within the REAL-AI-Benchmark approach: physical-AI benchmarking. Its core value is not that the task is large, but that the processing chain is complete. A model must see, read, assess, simulate, decide, and verify. This is much closer to real technical work than a task in which all numbers are already perfectly digitized.

The most important message of GO-6 is that intelligent decision-making is not the same as automatically reacting to the largest alarm. A system may have high fuzzy risk while the most aggressive action is still wrong because of secondary consequences. A model that understands this demonstrates a higher level of physical-system reasoning than a model that merely follows the first intuitive label.

For that reason, GO-6 is a logical continuation of the GO-1 through GO-5 set, but also the beginning of a new class of tests in which LLM/LMM systems must demonstrate not only formal accuracy, but also measured physical attention, causal discipline, and operational technical responsibility.

15. Limitations and expected extensions

The reference GO-6 instance is intentionally compact. It should not be interpreted as a complete industrial-control benchmark or as a safety-certified decision procedure. Its purpose is methodological: to expose whether a model can maintain a disciplined chain from analog perception to numerical inference, from inference to dynamic simulation, and from simulation to executable verification.

Future extensions can introduce additional instrument layouts, non-linear scales, pointer occlusion, multiple simultaneous sensor faults, saturation limits for physically impossible values such as negative flow, and audit variants with changed action parameters. Such extensions should preserve the same scoring philosophy: additive evaluation of clearly separated competencies, complete raw-log preservation, and strict distinction between multimodal and non-visual test variants.

Appendix A. Canonical GO-6 prompt formulation

```
GO-6 — Multimodal Physical-AI Benchmark

Five images of analog instruments are attached to this prompt:
PIC1.png — temperature T [°C]
PIC2.png — pressure P [kPa]
PIC3.png — flow F [L/min]
PIC4.png — vibration V [mm/s]
PIC5.png — pressure trend D = dP/dt [kPa/min]

First read the nominal values and reasonable reading intervals. Do not treat an analog instrument as a
perfect digital sensor. Then perform fuzzy inference, dynamic simulation of four actions across three
ticks, outcome classification, and selection of the uniquely best action. Finally write Zig code that
reproduces the procedure and provide RESULTS_JSON.

Mandatory sections:
A-READING_ANALOG_INSTRUMENTS
B-PHYSICAL_CONSISTENCY
C-FUZZY_INFERENCE
D-DYNAMIC_SIMULATION
E-DECISION_AND_PROOF_OF_UNIQUENESS
F-ZIG_CODE
RESULTS_JSON
<END>
```

Appendix B. Reference RESULTS_JSON


```
{
  "sensor_readings": {
    "T_C": 68.5,
    "P_kPa": 132.0,
    "F_L_min": 4.2,
    "V_mm_s": 5.6,
    "dPdt_kPa_min": 2.1
  },
  "reading_intervals": {
    "T_C": [68.0, 69.0],
    "P_kPa": [131.0, 133.0],
    "F_L_min": [4.0, 4.4],
    "V_mm_s": [5.4, 5.8],
    "dPdt_kPa_min": [1.9, 2.3]
  },
  "aggregated_risk": 0.85,
  "initial_fuzzy_decision": "EMERGENCY_MODE",
  "dynamic_outcomes": {
    "MONITOR": "catastrophic",
    "DERATE": "unstable",
    "CONTROLLED_RELIEF": "unstable",
    "EMERGENCY_MODE": "catastrophic"
  },
  "recommended_action": "CONTROLLED_RELIEF",
  "recommended_action_index": 2,
  "unique_best_action": true
}
```


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